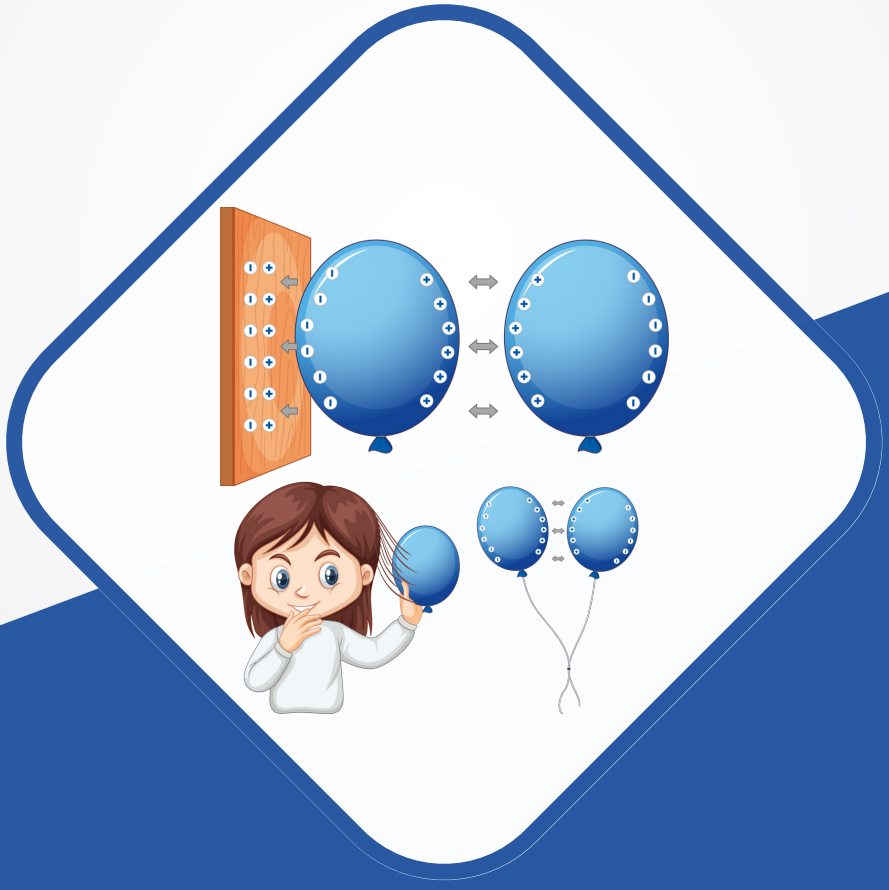
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Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Electrostatics

English Medium

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ELECTROSTATICS

KEY CONCEPT

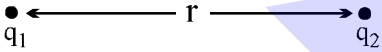
1. ELECTRIC CHARGE

Charge of a material body is that property due to which it interacts with other charges. There are two kinds of charges-positive and negative. S.I. unit is coulomb. Charge is quantized, conserved, and additive.

2. COULOMB'S LAW

$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$. In vector form $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^3}\vec{r}$

where ε_0 = permittivity of free space = 8.85×10^{-12} $N^{-1} m^{-2} c^2$ or F/m



Note : The Law is applicable only for static and point charges.
Moving charges may result in magnetic interaction. And if charges are extended, induction may change the charge distribution.

3. PRINCIPLE OF SUPERPOSITION

Force on a point charge due to many charges is given by $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ...$

Note : The force due to one charge is not affected by the presence of other charges.

4. ELECTRIC FIELD, ELECTRIC INTENSITY OR ELECTRIC FIELD STRENGTH (VECTOR QUANTITY)

In the surrounding region of a charge there exist a physical property due to which other charge experiences a force. The direction of electric field is direction of force experienced by a positively charged particle and the magnitude of the field (electric field intensity) is the force experienced by a unit charge.

$\vec{E} = \frac{\vec{F}}{q}$ unit is N/C or V/m.

5. ELECTRIC FIELD LINES (EFL)

Electric field lines are pictorial representation of electric field. They are drawn in such a way that tangent at any point is in the direction of electric field at that point. Therefore an arrow is put on electric field line pointing away from positive charge and towards the negative charge.

Properties of (EFL) :

(i) Electric field lines never intersect but they can touch each other.

(ii) EFL originate from positive charge and terminate on a negative charge. If an EFL is originated, it must require termination either at a negative charge or at ∞.

(iii) In a charge free region electric field lines are continuous.

(iv) Number of EFL originated or terminated from a charge or on a charge is proportional to the magnitude of charge.

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6. ELECTRIC FLUX

(i) For uniform electric field; $\phi = \vec{E}.\vec{A} = EA\cos\theta$ where θ = angle between $\vec{E}$ & area vector $(\vec{A})$.
Flux is contributed only due to the component of electric field which is perpendicular to the plane.

(ii) If $\vec{E}$ is not uniform throughout the area A, then $\phi = \int \vec{E}.d\vec{A}$

7. GAUSS'S LAW

(Applicable only to closed surface) Net flux emerging out of a closed surface is $\frac{q_{en}}{\varepsilon_0}$.

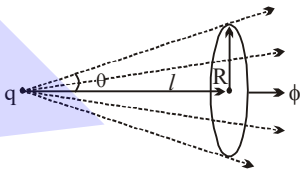
$$\phi = \oint \vec{E}\cdot d\vec{A} = \frac{q_{en}}{\varepsilon_0}$$

q = net charge enclosed by the closed surface .

ϕ does not depend on the
(i) Shape and size of the closed surface
(ii) The charges located outside the closed surface.

Concept of solid angle :
Flux of charge q having through the circle of radius R is

$$\phi = \frac{q/\in_0}{4\pi} \times \Omega = \frac{q}{2\in_0}(1-\cos\theta)$$



Solid angle of cone of half angle θ is $\Omega=2\pi(1-\cos\theta)$

8. ELECTRIC FIELD DUE TO

(i) Point charge : $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r} = \frac{1}{4\pi\in_0}\frac{q}{r^3}\vec{r}$ (vector form)



Where $\vec{r}$ = vector drawn from the source charge to the point value of q must be substituted with sign.

(ii) Continuous charge distribution $\vec{E} = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r^2}\hat{r} = \int d\vec{E}$

$d\vec{E}$ = electric field due to an elementary charge.

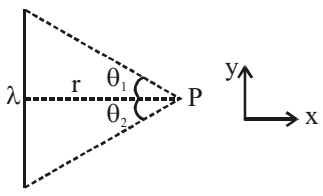
Note : $E \neq \int dE$ because E is a vector quantity.

$dq = \lambda d\ell$ (for line charge) = σds (for surface charge) = ρdv (for volume charge)
In general λ, σ & ρ are linear, surface and volume charge densities respectively.

(iii) Finite line charge

$$E_x = \frac{k\lambda}{r}[\sin\theta_1 + \sin\theta_2]$$

$$E_y = \frac{k\lambda}{r}[\cos\theta_2 - \cos\theta_1]$$



(iv) Infinite line of charge $|\vec{E}| = \frac{2k\lambda}{r}$ where r = perpendicular distance of the point from the line charge.

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(v) Semi infinite line of charge $|\vec{E}| = \frac{\sqrt{2}k\lambda}{r}$ as, $E_x = \frac{k\lambda}{r}$ & $E_y = \frac{k\lambda}{r}$ at a point above the end of wire at an angle 45°.

(vi) Uniformly charged ring, $E_{centre} = 0$, $E_{axis} = \frac{K\,Q\,x}{(x^2 + R^2)^{3/2}}$

Electric field is maximum when $\frac{dE}{dx} = 0$ for a point on the axis of the ring. Here we get $x = R/\sqrt{2}$.

(vii) Infinite non conducting sheet of charge $\vec{E} = \frac{\sigma}{2\varepsilon_0}\hat{n}$ where $\hat{n}$ = unit vector normal to the plane of sheet, where σ is surface charge density.

(viii) Uniformly charged solid sphere (Insulating material)

$E_{out} = \frac{Q}{4\pi\varepsilon_0 r^2}$; r ≥ R , Behaves as a point charge situated at the centre for these points

$E_{in} = \frac{Qr}{4\pi\varepsilon_0 R^3} = \frac{\rho r}{3\varepsilon_0}$; $r \le R$ where ρ = volume charge density

(ix) Uniformly charged cylinder with a charge density ρ is $E_{in} = \frac{\rho r}{2\in_0}$; for $r < R$

$E_{out} = \frac{\rho R^2}{2\in_0 r}$; for $r > R$

(x) Uniformly charged cylindrical shell with surface charge density σ is

$E_{in} = 0$; for $r < R$ $E_{out} = \frac{\rho r}{\in_0 r}$; for $r > R$

9. ELECTRIC POTENTIAL (Scalar Quantity)

Work done by external agent to bring a unit positive charge (without acceleration) from infinity to a point in an electric field is called electric potential at that point. If $W_{\infty r}$ is the work done to bring a charge q (very small) from infinity to a point then potential at that point is $V = \frac{(W_{\infty r})_{ext}}{q}$; S.I. unit is volt (=1 J/C)

- Potential at a point due to positive charge is positive & due to negative charge is negative.

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10. POTENTIAL DIFFERENCE

$$V_{AB} = V_A - V_B = \frac{(W_{BA})_{ext}}{q}$$

V_{AB} = potential difference between point A & B.
W_{BA} = work done by external source to transfer a point charge q from B to A (Without acceleration).

- Potential difference between two points in an electric field does not depend on the path between them.

11. ELECTRIC FIELD & ELECTRIC POTENTIAL

$\vec{E} = -\text{grad } V = -\vec{\nabla}V$ {read as gradient of V} grad $= \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$

Here electric potential varies with x, y and z coordinates.
For finding potential difference between two points in electric field, we use

$V_A - V_B = -\int_A^B \vec{E}.\overrightarrow{d\ell}$ if E is varying with distance

$= -Ed$ if E is constant & here d is the distance between points A and B.

- Positive charge flows from higher to lower (i.e. in the direction of electric field) potential and negative charge from lower to higher (i.e. opposite to the electric field) potential .

12. POTENTIAL DUE TO

(i) A point charge $V = \frac{Q}{4\pi\varepsilon_0 r}$

(ii) Many charges $V = \frac{q_1}{4\pi\varepsilon_0 r_1} + \frac{q_2}{4\pi\varepsilon_0 r_2} + \frac{q_3}{4\pi\varepsilon_0 r_3} + ...$

(iii) Continuous charge distribution $V = \frac{1}{4\pi\varepsilon_0}\int\frac{dq}{r}$

(iv) Uniformly charged spherical shell

$V_{out} = \frac{Q}{4\pi\varepsilon_0 r}$; $(r \geq R)$, $V_{in} = \frac{Q}{4\pi\varepsilon_0 R}$; $(r \leq R)$

(v) Non conducting uniformly charged solid sphere :

$V_{out} = \frac{Q}{4\pi\varepsilon_0 r}$; $(r \geq R)$, $V_{in} = \frac{1}{2}\frac{Q(3R^2 - r^2)}{4\pi\varepsilon_0 R^3}$; $(r \leq R)$

13. EQUIPOTENTIAL SURFACE

In an electric field the locus of points of equal potential is called an equipotential surface. An equipotential surface and the electric field meet at right angles. The region where $E = 0$, Potential of the whole region must remain constant as no work is done in displacement of charge in it.

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14. MUTUAL POTENTIAL ENERGY OR INTERACTION ENERGY

"The work to be done to integrate the charge system".

For 2 particle system $U_{\text{mutual}} = \dfrac{q_1 q_2}{4\pi\varepsilon_0 r}$

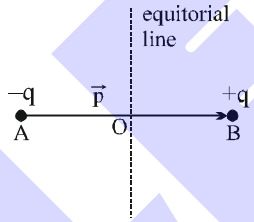
For 3 particle system $U_{\text{mutual}} = \dfrac{q_1 q_2}{4\pi\varepsilon_0 r_{12}} + \dfrac{q_2 q_3}{4\pi\varepsilon_0 r_{23}} + \dfrac{q_3 q_1}{4\pi\varepsilon_0 r_{31}}$

For n particles there will be $\dfrac{n\,(n-1)}{2}$ terms.

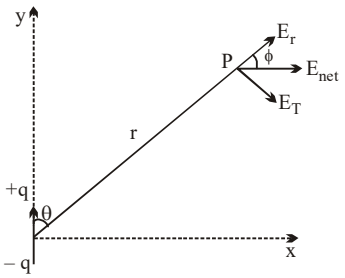
Potential energy of charge q in potential field $U = qV$
Interaction energy of a system of two charges $U = q_1 V_2 = q_2 V_1$

15. ELECTRIC DIPOLE

- Dipole moment $\vec{p} = q\vec{d}$
  (d is the separation between the charges and from $-q$ to $+q$)



- Electric field at a general point $P(r, \theta)$ in polar coordinate system is



Radial electric field $E_r = \dfrac{2Kp\cos\theta}{r^3}$

Tangential electric field $E_T = \dfrac{Kp\sin\theta}{r^3}$

Net electric field at P is $E_{net} = \sqrt{E_r^2 + E_T^2} = \dfrac{kp}{r^3}\sqrt{1+3\cos^2\theta}$

$\tan\phi = \dfrac{1}{2}\tan\theta$

- Electric Potential at point P is $V_P = \dfrac{Kp\cos\theta}{r^2} = \dfrac{\vec{p}\cdot\vec{r}}{4\pi\varepsilon_0 r^3}$
- Electric Dipole in uniform electric field : Torque $\vec{\tau} = \vec{p} \times \vec{E}$; $\vec{F} = 0$.
- Work done in rotation of dipole is $W = pE\,(\cos\theta_1 - \cos\theta_2)$
- Potential energy of an electric dipole in electric field $U = -\vec{p}\cdot\vec{E}$.

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- When $\vec{p} \parallel \vec{E}$ the dipole is in stable equilibrium
- When $\vec{p} \parallel (-\vec{E})$ the dipole is in unstable equilibrium

16. CONDUCTORS

- In electrostatic condition, the net electric field inside the volume of the conductor is zero.
- When a conductor is charged, the charge resides only on the surface.
- Electric field is always perpendicular to a conducting surface (or any equipotential surface). No tangential component of electric field on such surfaces.
- For a conductor of any shape E (just outside) = $\frac{\sigma}{\varepsilon_o}$
- Charge density at convex sharp points on a conductor is greater. Lesser is radius of curvature at a convex part, greater is the charge density.
- Infinite charged conducting sheet having surface charge density σ on each of surfaces $E = \sigma/\varepsilon_o$.
- Just outside a conducting surface charged with a surface charge density σ, electric field is always given as $E = \sigma/\varepsilon_o$.
- Uniformly charged spherical shell (conducting) or uniformly charged solid conducting sphere.

$E_{out} = \frac{Q}{4\pi\varepsilon_0 r^2}$; $r \geq R$ Behaves as a point charge situated at the centre for these points.

$E_{in} = 0$; $r < R$

17. ELECTROSTATICS PRESSURE AND ENERGY DENSITY

- In vacuum energy stored per unit volume in an electric field = $\frac{\varepsilon_o E^2}{2}$
- Electric pressure due to its own charge on a surface having charged density σ is $P_{ele} = \frac{\sigma^2}{2\varepsilon_o}$.
- Electric pressure on a charged surface with charged density σ due to external electric field is $P_{ele} = \sigma E_1$

18. SELF POTENTIAL ENERGY

Total energy of a system = $U_{self} + U_{mutual}$

Self potential energy of a insulating uniformly charged spherical shell = $\frac{KQ^2}{2R}$.

Self potential energy of a charged conducting spherical shell = $\frac{KQ^2}{2R}$.

Self potential energy of an insulating uniformly charged sphere = $\frac{3KQ^2}{5R}$.

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